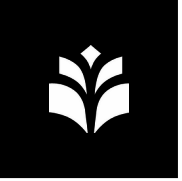


Intro to the web

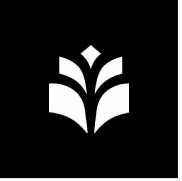
Web programming I

Course materials originally by Sonsoles López-Pernas, adapted by Maria Kuismin



Intro to the web

Student is able to
identify and explain the principles of client–server architecture and the HTTP protocol.



Content

Internet ja web

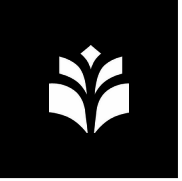
- What is the difference between the Internet and the Web?
- How did the Web evolve over time?

Client–server architecture

- What is the client-server architecture?

HTTP

- What is HTTP, and how does it work?
- What HTTP methods and response codes do we need to know?



Internet ja web

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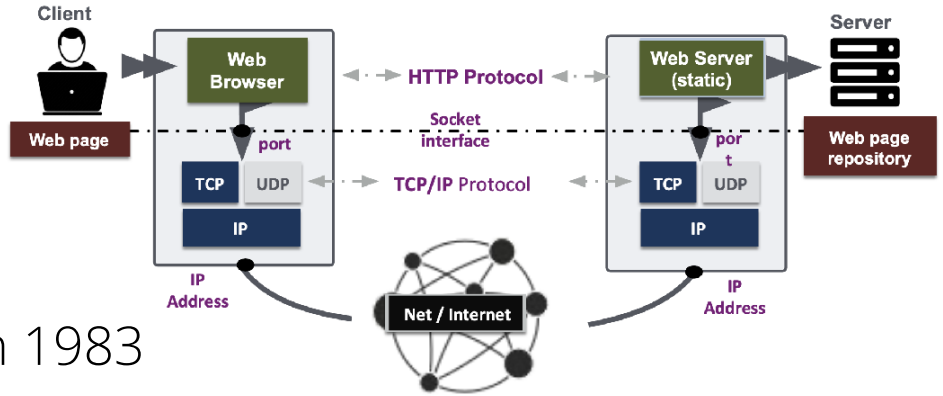
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Origin of the Internet



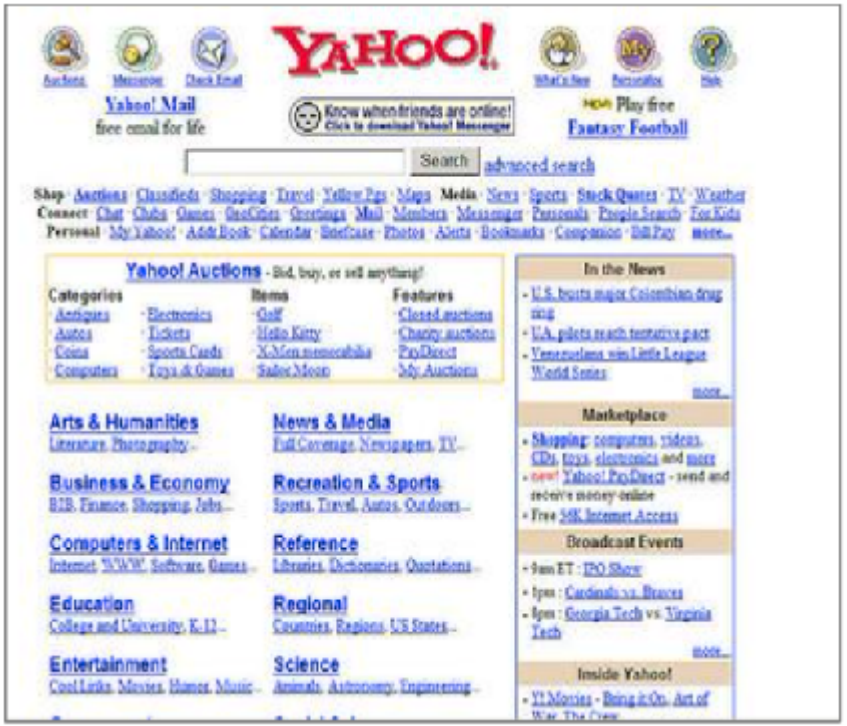
- The Internet started as ARPANET in 1983
- The Internet connects computers to the network using the TCP/IP protocol
- TCP/IP supports client-server applications through a socket interface
  - The client-server model is a distributed application structure that divides the tasks to be performed between resource or service providers, called servers, and those interested in these services or products, called clients.
- The first applications on the Internet were: *telnet* (virtual terminal), *ftp* (file transfer), *email*, ...

WWW – the Web

- Tim Berners Lee proposed a new application in 1989: the World Wide Web
- The Web is an information space where we find hypertext documents and other resources uniquely identified by their Uniform Resource Locator (URL), interconnected by hyperlinks and accessible through the Internet.
- There is a client application (text browser) and a server application (static web server)
- Tim Berners Lee developed/joined three essential technologies:
  - URL: Uniform Resource Locator – address of a resource
  - HTML: Hypertext Markup Language – structure of content
  - HTTP: Hypertext Transfer Protocol – communication protocol

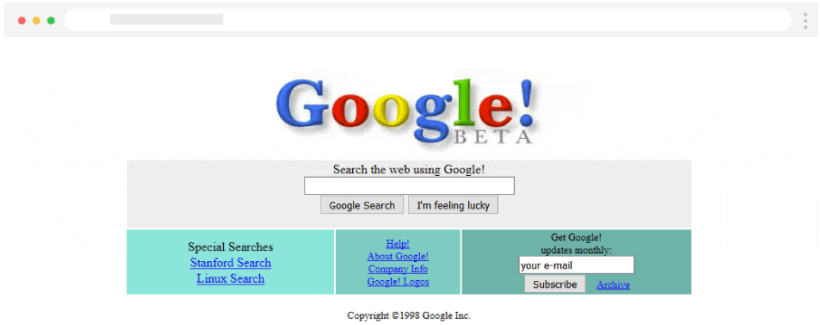
The Web in the 90s

- **Static** pages are served (HTTP GET) and forms are sent (HTTP POST).
- With each request, all displayed content is **refreshed**: complete pages.
- Logic is only on the **server side**.



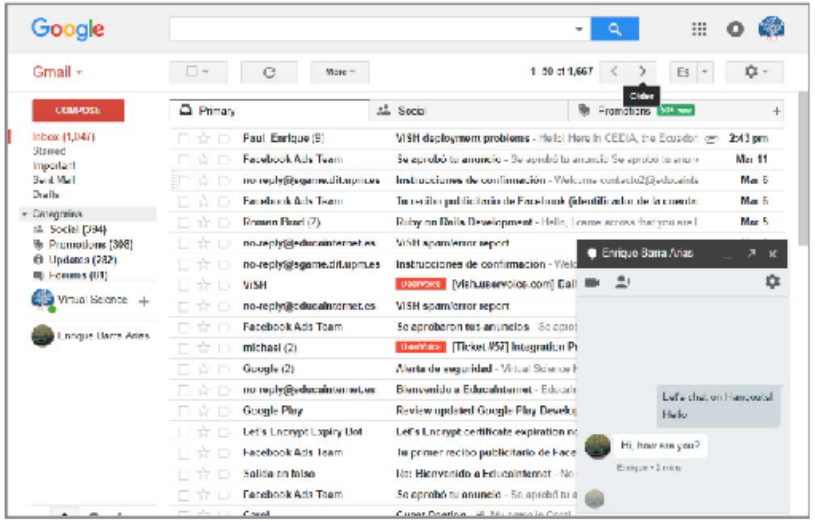
Dynamic Web in late 1990s

- Pages were **generated** on the server per request
- Often using PHP or ASP, with cookies and databases enabling **user-specific customization**.
- Each interaction still caused a **full-page reload**, but what appeared was increasingly personalized and responsive.

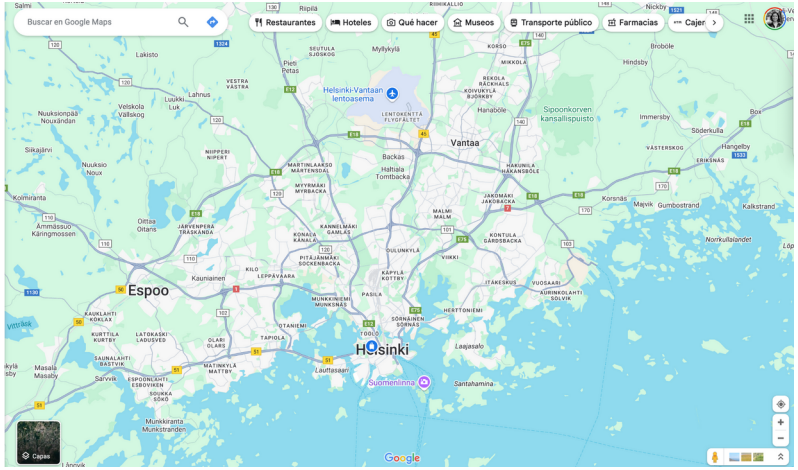


The Web in the 2000s

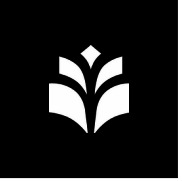
- Fetching **pieces** of pages without reloading the entire page.
- **AJAX** (Asynchronous JavaScript and XML) - XMLHttpRequest
- Use of **JavaScript** to communicate asynchronously with the server.
- Most of the logic is on the server.



Today’s Web



- **SPA** (Single Page Application)
- Web application architecture. Similar to desktop applications.
- Most of the logic is on the **client**.
- The server is only for storing data and authentication and not much else.
- With the first request the browser fetches the web application, and later other technologies are used to communicate with the server and send and receive data from the server.
  - AJAX, Websockets, RTSP, WebRTC, …



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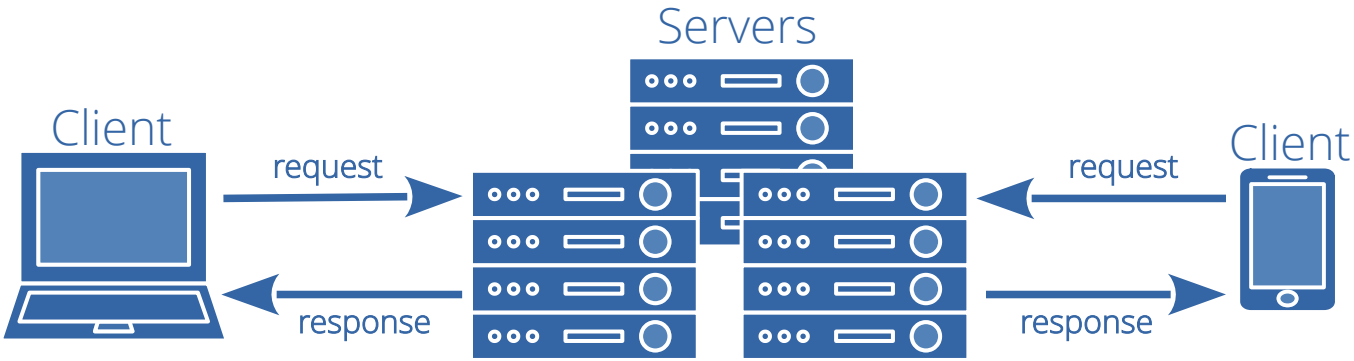
Client–server architecture

- What is the client-server architecture?

HTTP

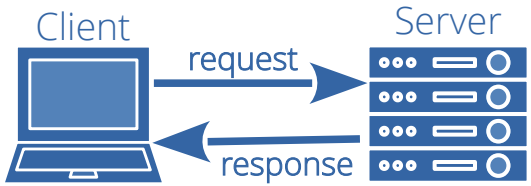
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Overview



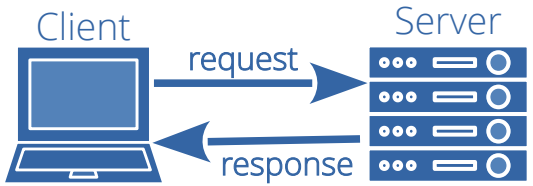
The Client makes request for service to server.
The service responds to the request.

Servers



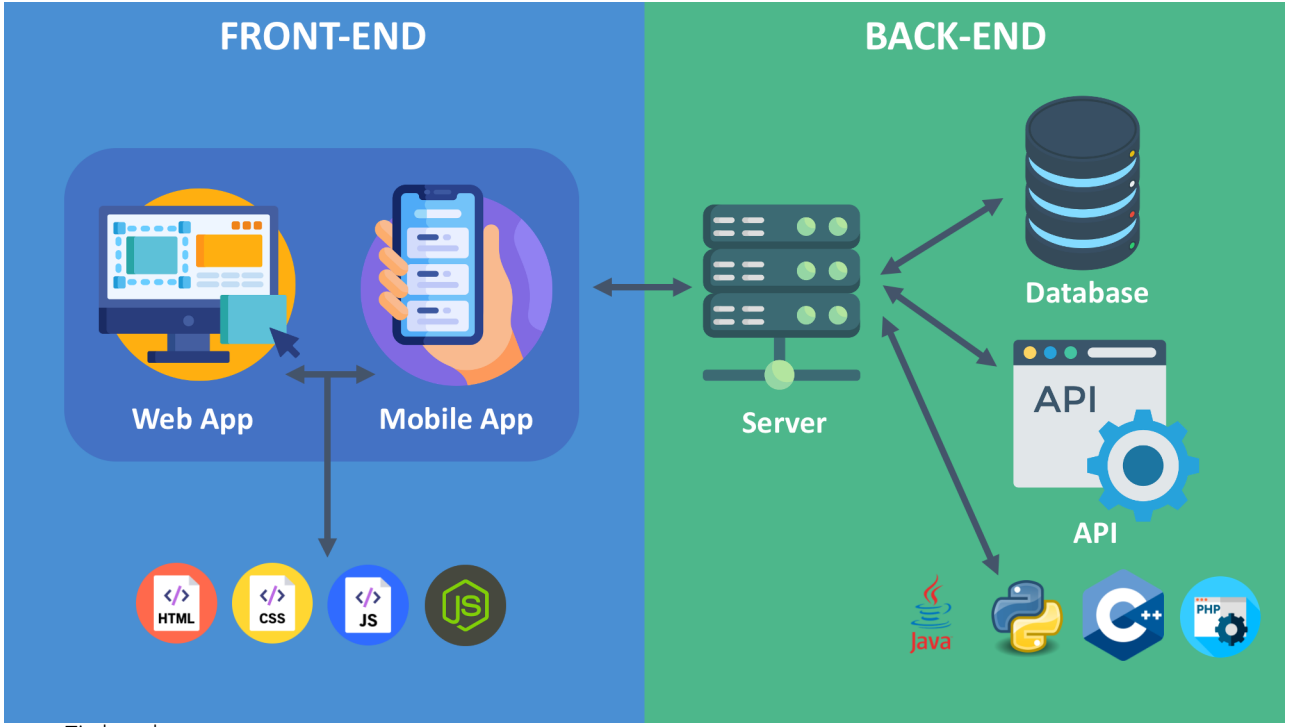
- **machine** that serves the content
  - a physical machine or a virtual machine in the cloud
  - server machine has a “known” address on the Internet (DNS name resolution-> IP)
- **program** that provides services to clients is also called a server.
  - Server programs run on a server machine.
    - Each program connects to a port on the server machine (the Web server uses port 80 by default).
  - Examples of web servers: Apache, Nginx, Microsoft-IIS, Nodejs, etc.

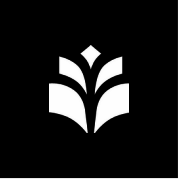
Clients



- **devices** that connect to the Internet: PCs, laptops, tablets, phones, etc.
- **programs** that access services on the Internet
  - **browser** is the main access client: Chrome, Firefox, Opera, Safari, ...
    - content served on the browser is programmed in HTML, CSS and JavaScript.
  - **mobile** device apps are the most used clients today.
    - Native applications (apps): Android, iOS, etc.
    - programmed in development environments with specific languages: Android programmed in Java/Kotlin, iOS with Swift, etc.

Client-server on web development





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Client–server architecture

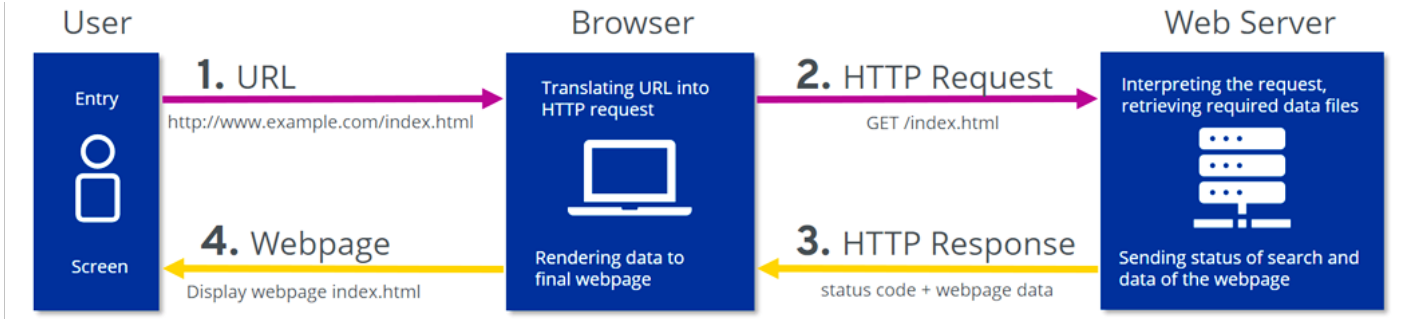
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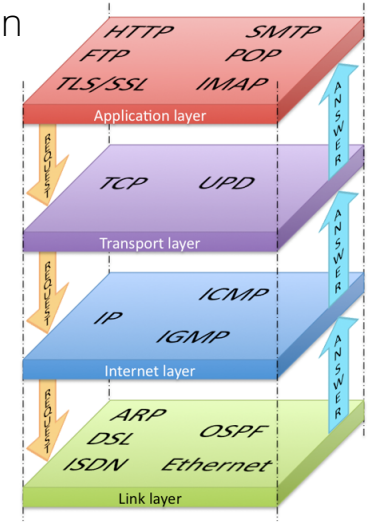
HTTP

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Hypertext Transfer Protocol

- supports hypermedia access to Web pages and other Internet services
- stateless transactional request-response protocol
  - HTTP transaction is always initiated from the client side
- the client establishes HTTP connection with the server or reuses an existing one
- the client sends HTTP Requests to the server over the HTTP connection
- the server answers the client by sending HTTP Responses over the connection
  - request and response are independent, though related, transmissions
  - the request and response have the same format





HTTP methods

The type of primitive of an HTTP request is indicated by the method or verbs:

- Uniform interface, methods allow you to manage resources on the server with the four CRUD primitives:
  - **POST**: Create a resource on the server.
  - **GET**: Read a resource from the server.
  - **PUT**: Update a resource from the server.
  - **DELETE**: Delete a resource from the server.

HTTP Responses

Informative Responses (1xx)

- 100 Continue: Continue partial request

Request completed (2xx)

- 200 OK: Operation completed successfully, resource served
- 201 Created: Resource created or updated successfully

Redirect (3xx)

- 301 Moved Permanently: Resource on another server
- 304 Not Modified: When the client already has the data

Client Error (4xx)

- 400 Bad Request: Incorrect command sent
- 404 Not Found: Resource not found in repository
- 405 Method Not Allowed: Method not supported by server
- 409 Conflict: Conflict with the resource status
- 410 Gone: Resource no longer exists

Server Error (5xx)

- 500 Internal Server Error: The server has errors

MIME type (Multipurpose Internet Mail Extensions)

- defines the type of a resource or file in email, Web, ...
- format: "type/subtype":
  https://developer.mozilla.org/ar/docs/Web/HTTP/Basics_of_HTTP/MIME_Types
- most common MIME types for the Web

– **Text type:**

text/plain	Plain text without format
text/html	HTML
text/css	CSS file
text/javascript	JavaScript script

– **Image type:**

image/gif	Image in GIF format
image/jpeg	Image in JPEG format
image/png	Image in PNG format
image/svg	Image in SVG vector format

– **Sending form parameters in the body:**

application/x-www-form-urlencoded Parameters encoded in URL-encoded

multipart/form-data Parameters with multiple formats

– **Server response with multiple blocks and parts:**

multipart/byte-ranges Response in multiple blocks

Quick check

Go to

www.menti.com

Enter the code

2651 9483



Or use QR code

Summary

- The Internet is the underlying network infrastructure, the global network of computers connected through TCP/IP; the Web is one application built on top of the Internet
- The Web has evolved from static pages, to dynamic server-rendered content, to Single Page Applications where most logic runs on the client.
- The client-server architecture divides tasks: clients (browsers, mobile apps) request services; servers provide content, data, and business logic.
- HTTP is a stateless, request-response protocol. The client always initiates the request, and the server responds.
- HTTP uses standard methods (GET, POST, PUT, DELETE) to manage resources, and response codes (1xx, 2xx, 3xx, 4xx, 5xx) to indicate the result.

Bibliography

- Materials of the course *Computación en Red* - Universidad Politécnica de Madrid